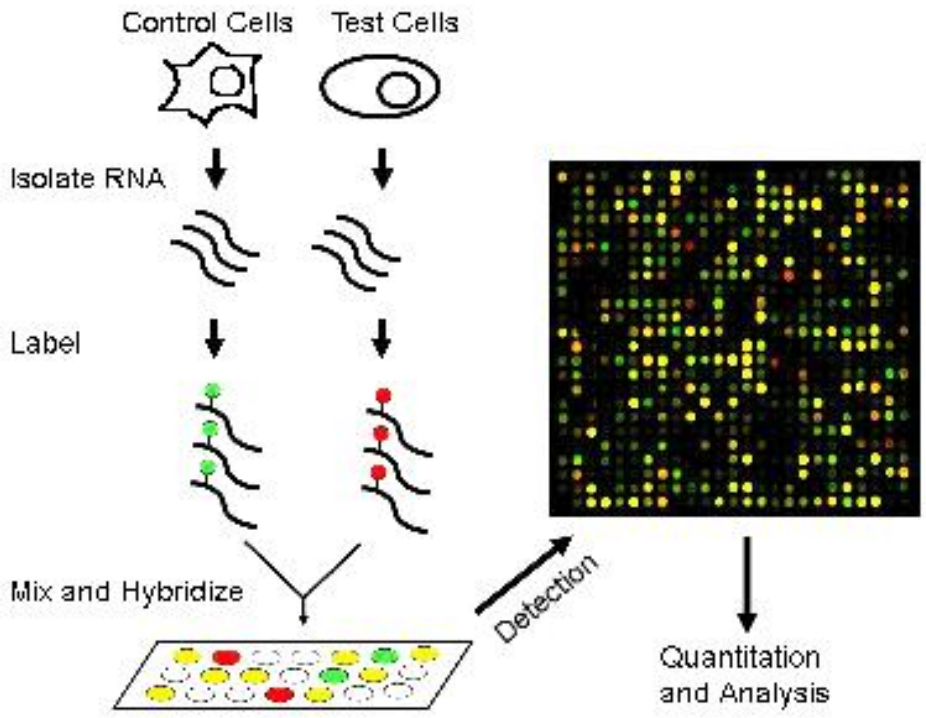
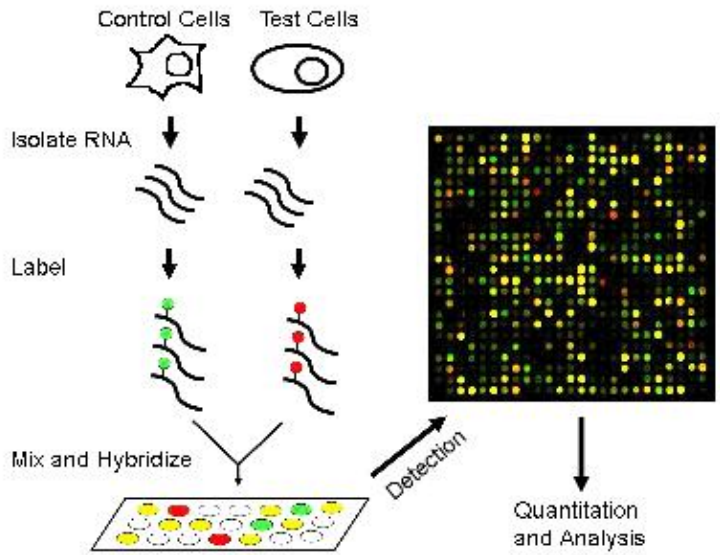
Lecture 16: Protein Expression



Microarrays, proteomics, protein interaction

Outline

1. **Protein Expression**
2. Correlations and Causality



Protein Interactions

- Cells are complex organisms where proteins work together to solve problems
- The concentration of proteins will vary depending on what the cell is doing
- The concentration of one protein will affect (excite or inhibit) the expression of other proteins
- This lecture is about try to find these interactions
- We will later look at the interactions in detail

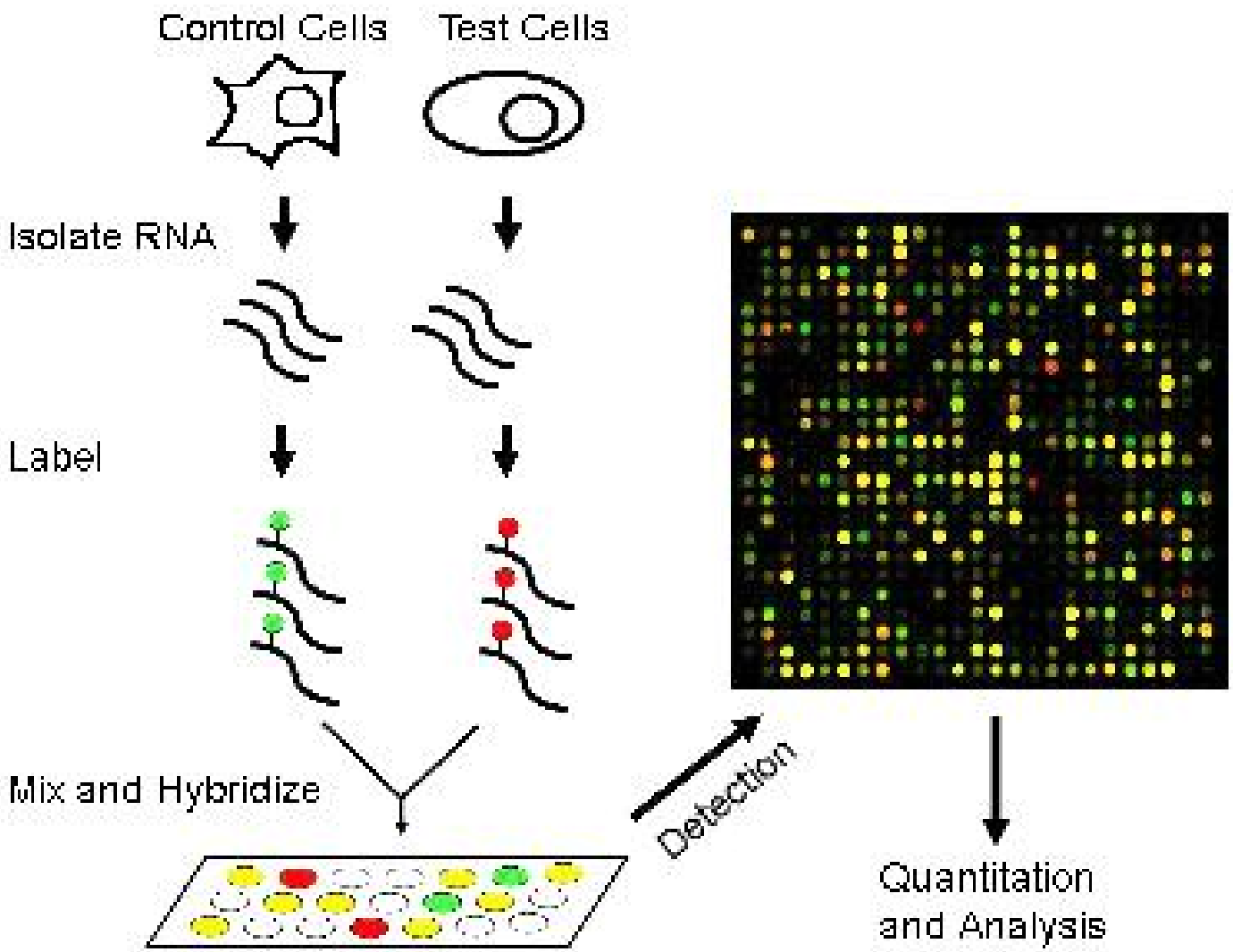
Proteomics

- The direct measure of protein concentration is known as **proteomics**
- It is a big subject in itself
- Chemically there is no easy way to read off a protein sequence
- We can measure the mass and charge of a molecule accurately using a mass spectrometer
- This is used to directly measure protein concentration

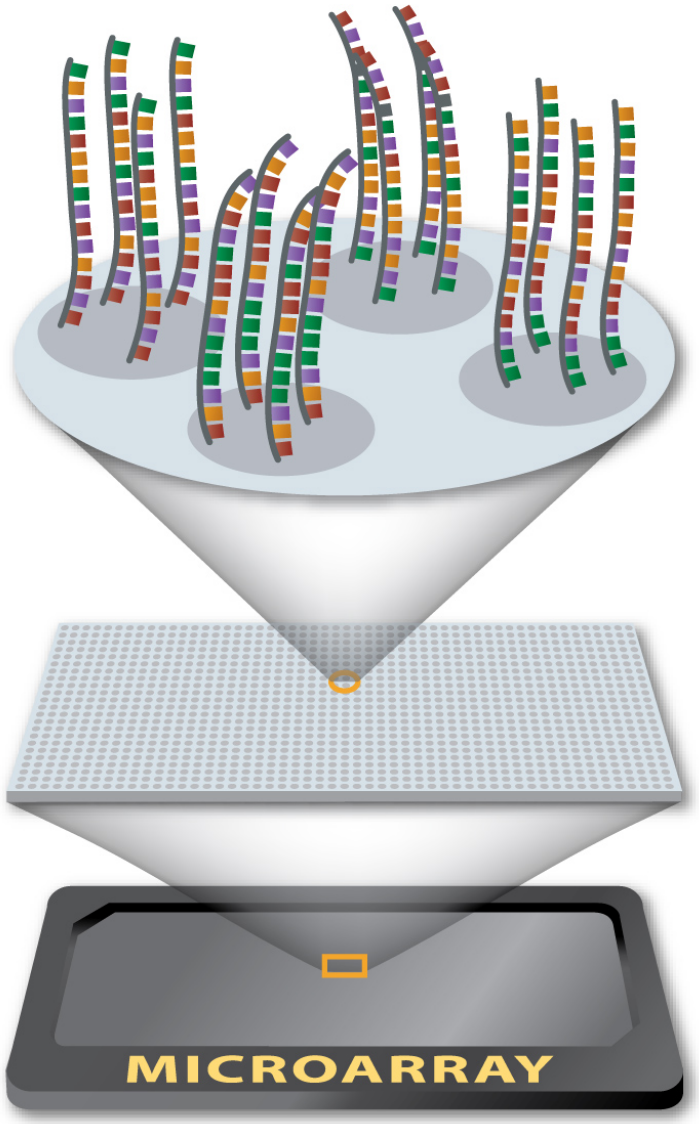
Measuring mRNA Levels

- An indirect method to measure protein concentration is to measure the concentration of mRNA
- Although not a perfect proxy it does provide a reasonable approximation to the protein concentration
- It is much easier and cheaper than direct measurement of protein concentration
- Two methods
  - ⋆ Microarrays (used for the last 20 years)
  - ⋆ RNA sequencing (beginning to take over)

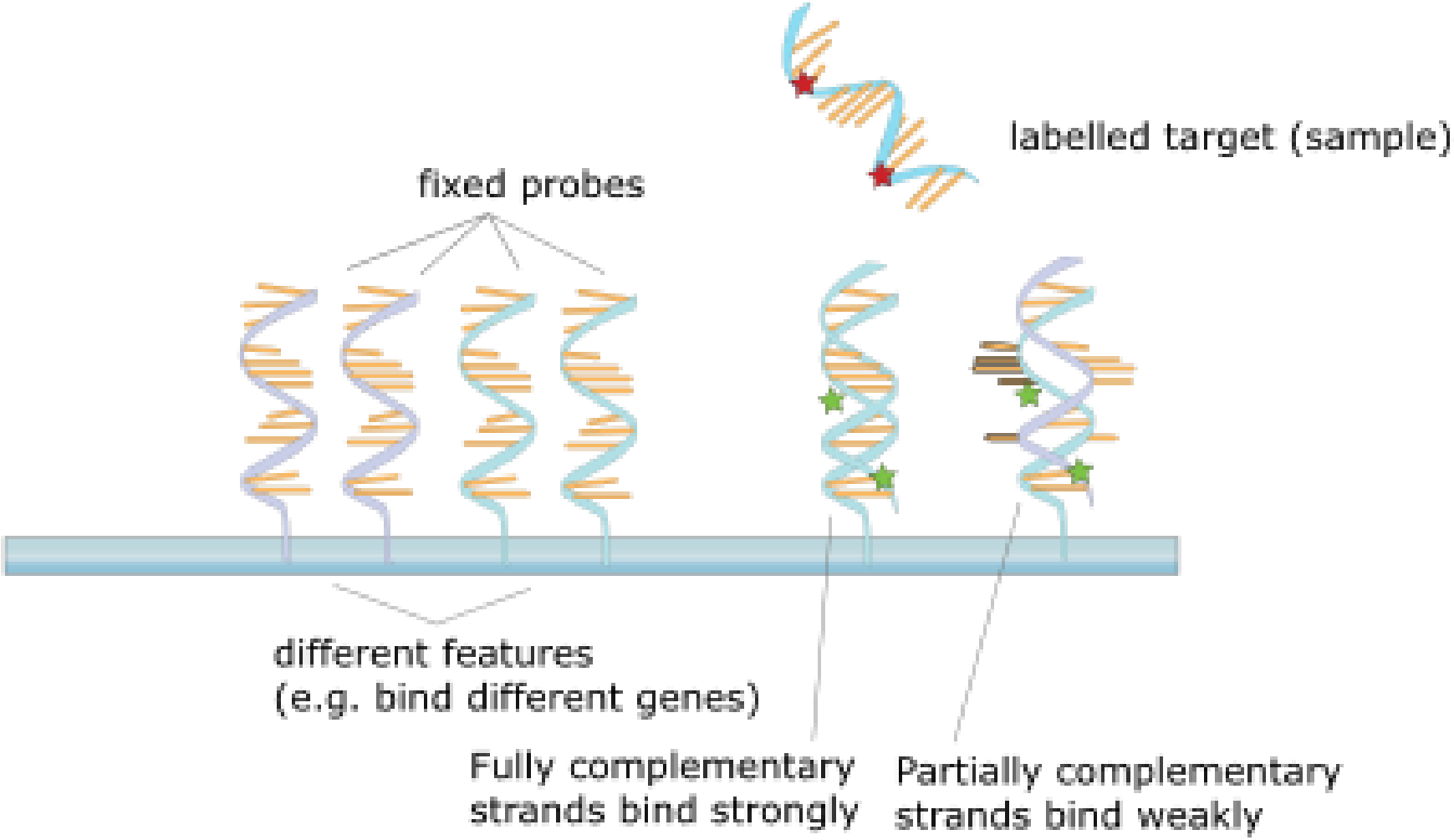
DNA Microarrays



DNA Microarrays



DNA Microarrays



DNA Microarrays

- Single stranded mRNA is synthesised to double stranded cDNA
- The cDNA will bind to a complementary strand attached to a microarray
- The microarray needs to by built with the correct DNA strands for the organism being used
- Typically a microarray will have 10 000–100 000 different DNA sequences
- Microarray technology is relatively cheap ($\approx$£100–£1000)

Using DNA Microarray

- All tests are carried out relative to a control cell
- Results show over-expression (red) or under-expression (green) relative to a group of control cells
- Study cell that is doing something different to the control
- Always use a population of cells so we are only looking at population averages

Time Series

- To get an understanding of the correlation between proteins we usually study the evolution over time of the cells
- E.g. add sugar, starve the cell, increase the temperature
- Note, that we are looking at a different population of cells since we kill the cells to extract RNA
- Cells will typically be dividing through the experiment
- Measurements are noisy by their nature

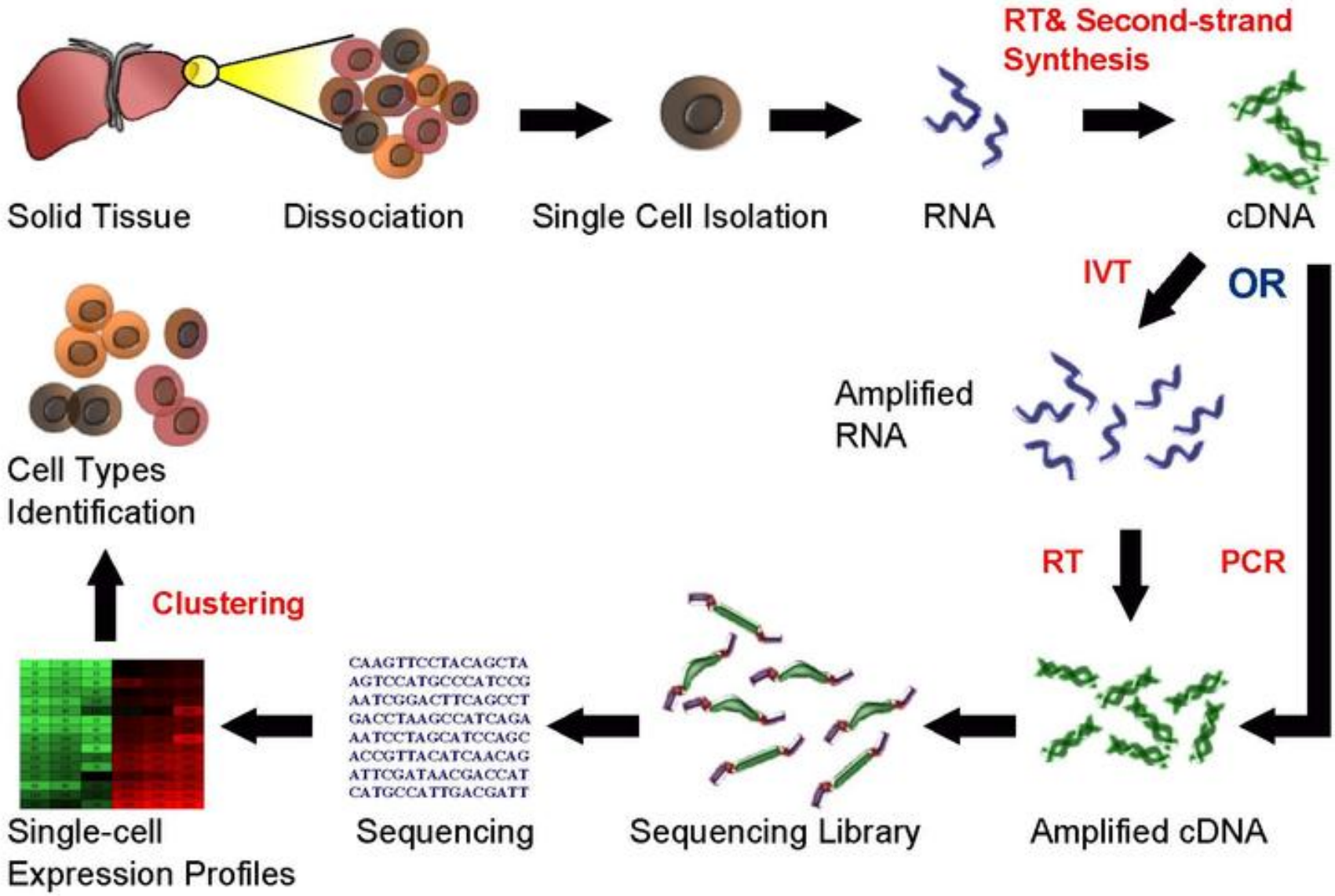
RNA-SEQ

- Recently, sequencing technology has become so cheap that it is feasible to sequence all the mRNAs *de novo*
- This prevents the need for having to tailor the microarray to a specific organism (can move away from model organisms)
- Microarrays can saturate and can provide unreliable counts for poorly expressed proteins due to background noise
- Will find rarely expressed RNA that might not be in microarray

Single Cell RNA-SEQ

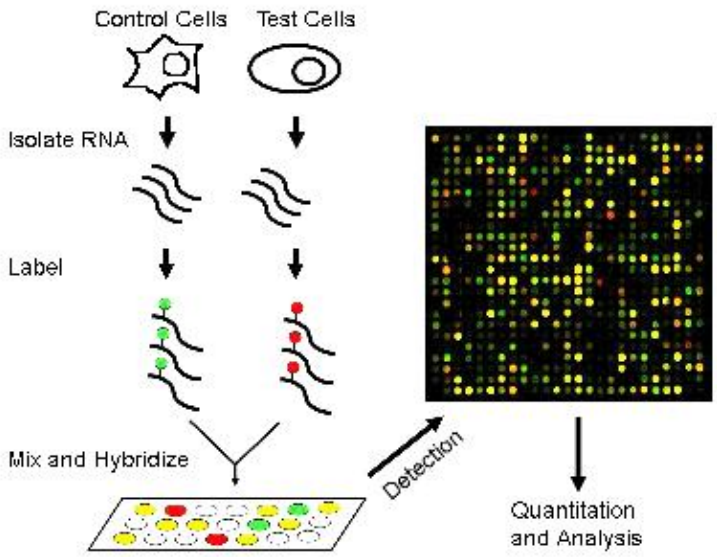
- With third generation sequencing technology it is even possible to measure the RNA concentrations in a single cell
- This is expensive, but allows researchers to identify different cells in a population
- Can be very important, for examples, in studying tumours
- Nevertheless, due to amplification processes, etc. there can be biases

Single Cell RNA Sequencing Workflow



Outline

1. Protein Expression
2. **Correlations and Causality**



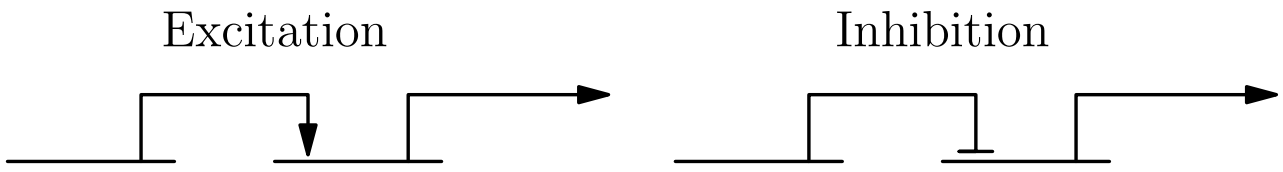
Correlation Between Time Series

- We suppose we have carried out an experiment where we have put a population of cells in a saline solution

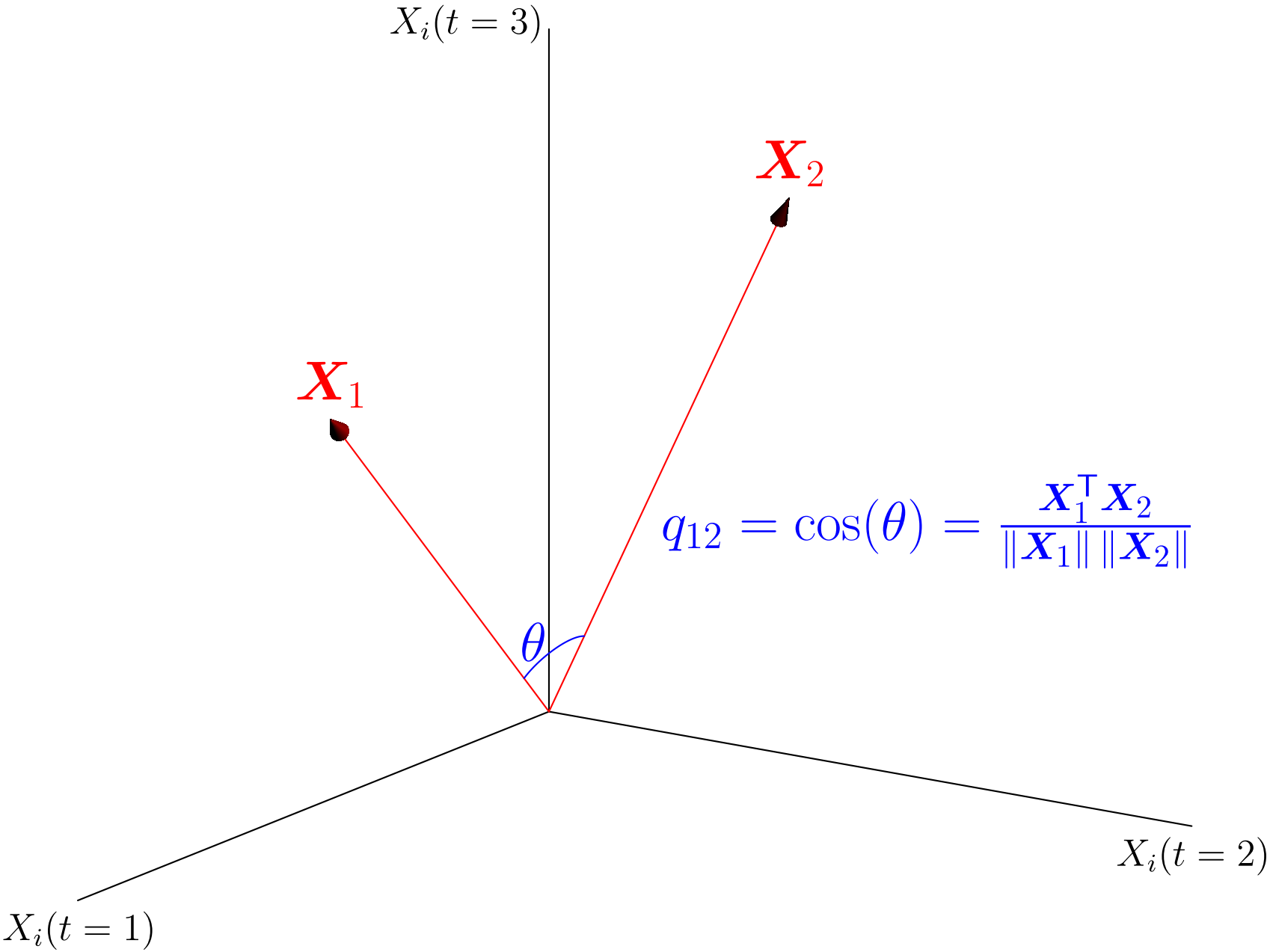
- For each mRNA we have a time series (e.g. taken every 30min)

$$\boldsymbol{X}_i = \big(X_i(1),\, X_i(2),\, \ldots,\, X_i(T)\big)$$

- We can immediately measure if two mRNAs are correlated or anti-correlated



Pearson Correlation



Correlation and Causality

- The famous mantra is

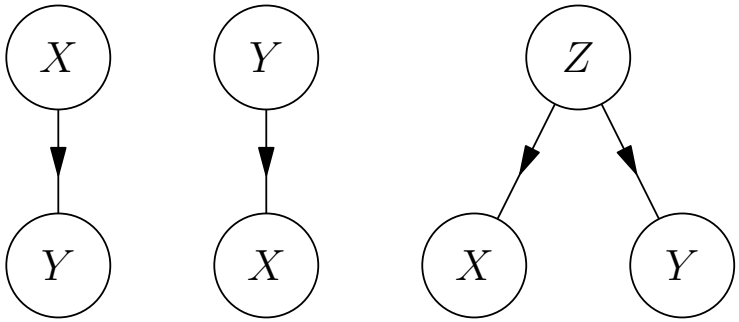
  Correlation does not imply causality

- If two random variables X and Y are correlated then
  - ⋆ X could affect Y
  - ⋆ Y could affect X
  - ⋆ X and Y could not influence each other, but both could be affected by another random variable Z

Graphical Models

- Graphical models are directed graphs that show causal relationships between random variables

- We could represent the three conditions described above by



- Want to infer a graphical model showing interactions between proteins

Statistical Independence

- Two random variables are statistically independent if

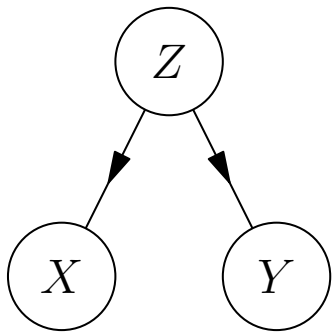
$$\mathbb{P}(X, Y) = \mathbb{P}(X)\,\mathbb{P}(Y)$$

- Equally this implies $\mathbb{P}(X|Y) = \mathbb{P}(X)$ and $\mathbb{P}(Y|X) = \mathbb{P}(Y)$
- Statistically independent variables are uncorrelated
- But statistical independence is often too powerful

Conditional Independence

- A weaker notion is conditional independence

$$\mathbb{P}(X, Y|Z) = \mathbb{P}(X|Z)\,\mathbb{P}(Y|Z)$$



- Conditional independence implies that there is no direct causation
- This is what we want to know
- But it doesn't imply zero correlation
- How do we identify conditionally independent quantities?

Multivariate Normal Distributions

- The simplest probability distribution for describing continuous variables is the multivariate normal distribution

$$\mathcal{N}(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}\sqrt{|\boldsymbol{\Sigma}|}}\,\mathrm{e}^{-\frac{1}{2}(\boldsymbol{X}-\boldsymbol{\mu})^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{\mu})}$$

- With

$$\mathbb{E}\left[\boldsymbol{X}\right] = \int \boldsymbol{X}\,\mathcal{N}(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma})\mathrm{d}\boldsymbol{X} = \boldsymbol{\mu}$$

$$\mathbb{C}\mathrm{ov}[\boldsymbol{X},\boldsymbol{X}] = \mathbb{E}\left[\boldsymbol{X}\,\boldsymbol{X}^{\mathsf{T}}\right] - \mathbb{E}\left[\boldsymbol{X}\right]\mathbb{E}\left[\boldsymbol{X}^{\mathsf{T}}\right] = \boldsymbol{\Sigma}$$

Covariance

- The covariance $\mathbb{C}\mathrm{ov}[X_i, X_j]$ measures how X_i varies with X_j
- The correlation is given by

$$q_{ij} = \frac{\mathbb{C}\mathrm{ov}[X_i, X_j]}{\sqrt{\mathbb{V}\mathrm{ar}[X_i]\ \mathbb{V}\mathrm{ar}[X_j]}}$$

- Where $\mathbb{V}\mathrm{ar}[X_i] = \mathbb{C}\mathrm{ov}[X_i, X_i] = \mathbb{E}\left[X_i^2\right] - \mathbb{E}\left[X_i\right]^2$ is the variance of X_i

Trivariant Normal Distribution

- Consider a 3-D normally distributed random variable $\boldsymbol{X} = (X_1,\, X_2,\, X_3)$

$$\mathcal{N}(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}\,|\boldsymbol{\Sigma}|}\exp\Bigg(-\tfrac{1}{2}\,\Sigma_{11}^{-1}(X_1-\mu_1)^2 - \tfrac{1}{2}\,\Sigma_{22}^{-1}(X_2-\mu_2)^2 - \tfrac{1}{2}\,\Sigma_{33}^{-1}(X_3-\mu_3)^2 - \Sigma_{12}^{-1}(X_1-\mu_1)\,(X_2-\mu_2) - \Sigma_{23}^{-1}(X_2-\mu_2)\,(X_3-\mu_3) - \Sigma_{13}^{-1}(X_1-\mu_1)\,(X_3-\mu_3)\Bigg)$$

- If $\Sigma_{13}^{-1} = 0$ then

$$\mathcal{N}(\boldsymbol{X}|\boldsymbol{\mu},\boldsymbol{\Sigma}) \propto \mathrm{e}^{-\frac{1}{2}\,\Sigma_{11}^{-1}(X_1-\mu_1)^2 - \Sigma_{12}^{-1}(X_1-\mu_1)\,(X_2-\mu_2)}\;\mathrm{e}^{-\frac{1}{2}\,\Sigma_{33}^{-1}(X_3-\mu_3)^2 - \Sigma_{23}^{-1}(X_2-\mu_2)\,(X_3-\mu_3)}\;\mathrm{e}^{-\frac{1}{2}\,\Sigma_{22}^{-1}(X_2-\mu_2)^2}$$

Conditional Independence

- If $\Sigma_{13}^{-1} = 0$ then

$$\mathcal{N}(\boldsymbol{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \propto \mathrm{e}^{-\frac{1}{2}\Sigma_{11}^{-1}(X_1-\mu_1)^2 - \Sigma_{12}^{-1}(X_1-\mu_1)(X_2-\mu_2)}$$
$$\mathrm{e}^{-\frac{1}{2}\Sigma_{33}^{-1}(X_3-\mu_3)^2 - \Sigma_{23}^{-1}(X_2-\mu_2)(X_3-\mu_3)}\ \mathrm{e}^{-\frac{1}{2}\Sigma_{22}^{-1}(X_2-\mu_2)^2}$$

- But $\mathbb{P}(X_1,\, X_3|X_2) = \mathbb{P}(X_1,\, X_2,\, X_3)/\mathbb{P}(X_2)$

- So

$$\mathbb{P}(X_1, X_3|X_2) = \mathbb{P}(X_1|X_2)\ \mathbb{P}(X_3|X_2)$$

- Thus $\Sigma_{13}^{-1} = 0 \Rightarrow X_1 \perp X_3|X_2$

Finding Conditional Independence

- If we suppose that the expression data at different times is normally distributed–which of course it isn't quite

- One way to find conditionally independent variables would be to measure the covariance between the RNA levels $\mathbb{C}\mathrm{ov}[X_i, X_j] = \Sigma_{ij}$ and invert the matrix $\boldsymbol{\Sigma}$

- But we can only estimate the empirical covariance

$$\hat{\Sigma}_{ij} = \frac{1}{T-1} \sum_{t=1}^{T} \left(X_i(t) - \hat{\mu}_i\right) \left(X_j(t) - \hat{\mu}_j\right), \quad \hat{\mu}_i = \frac{1}{T} \sum_{t=1}^{T} X_i(t)$$

- How do we tell if Σ_{ij}^{-1} is zero or just small?

Maximum Likelihood

- Given our data $\mathcal{D} = \left(X_i(t) \middle| i = 1, 2, \ldots, n \quad t = 1, 2 \ldots, T\right)$ we can try to find the value of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ that maximises the likelihood of the data

$$\mathbb{P}\left(\mathcal{D} | \boldsymbol{\mu}, \boldsymbol{\Sigma}\right) = \prod_{t=1}^{T} \mathcal{N}(\boldsymbol{X}(t) | \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$= \left(\frac{1}{(2\pi)^{d/2} \left|\boldsymbol{\Sigma}\right|^{1/2}}\right)^{T} \prod_{t=1}^{T} \mathrm{e}^{-\frac{1}{2}\left(\boldsymbol{X}(t) - \boldsymbol{\mu}\right)^{\mathsf{T}} \boldsymbol{\Sigma}^{-1} \left(\boldsymbol{X}(t) - \boldsymbol{\mu}\right)}$$

- Where $\boldsymbol{X}(t) = \left(X_1(t), X_2(t), \ldots, X_n(t)\right)$ (note that when looking at correlation we considered $\boldsymbol{X}_i$ as vectors)

Maximum Log-Likelihood

- The maximum of the likelihood will also be the maximum of the log-likelihood $\mathcal{L} = \log(\mathbb{P}\left(\mathcal{D}|\boldsymbol{\mu},\, \boldsymbol{\Sigma}\right))$

$$\mathcal{L} = -\frac{T}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}\sum_{t=1}^{T}(\boldsymbol{X}(t) - \boldsymbol{\mu})^{\mathsf{T}}\,\boldsymbol{\Sigma}^{-1}\,(\boldsymbol{X}(t) - \boldsymbol{\mu}) + C$$

- If we maximise with respect to $\boldsymbol{\mu}$ (i.e. solving $\boldsymbol{\nabla}_{\boldsymbol{\mu}}\mathcal{L} = 0$) we find $\boldsymbol{\mu}^* = \hat{\boldsymbol{\mu}}$

- Using the maximum likelihood mean

$$\mathcal{L} = -\frac{T}{2}\log(|\boldsymbol{\Sigma}|) - \frac{1}{2}\mathrm{tr}\,\mathbf{S}\,\boldsymbol{\Sigma}^{-1}$$

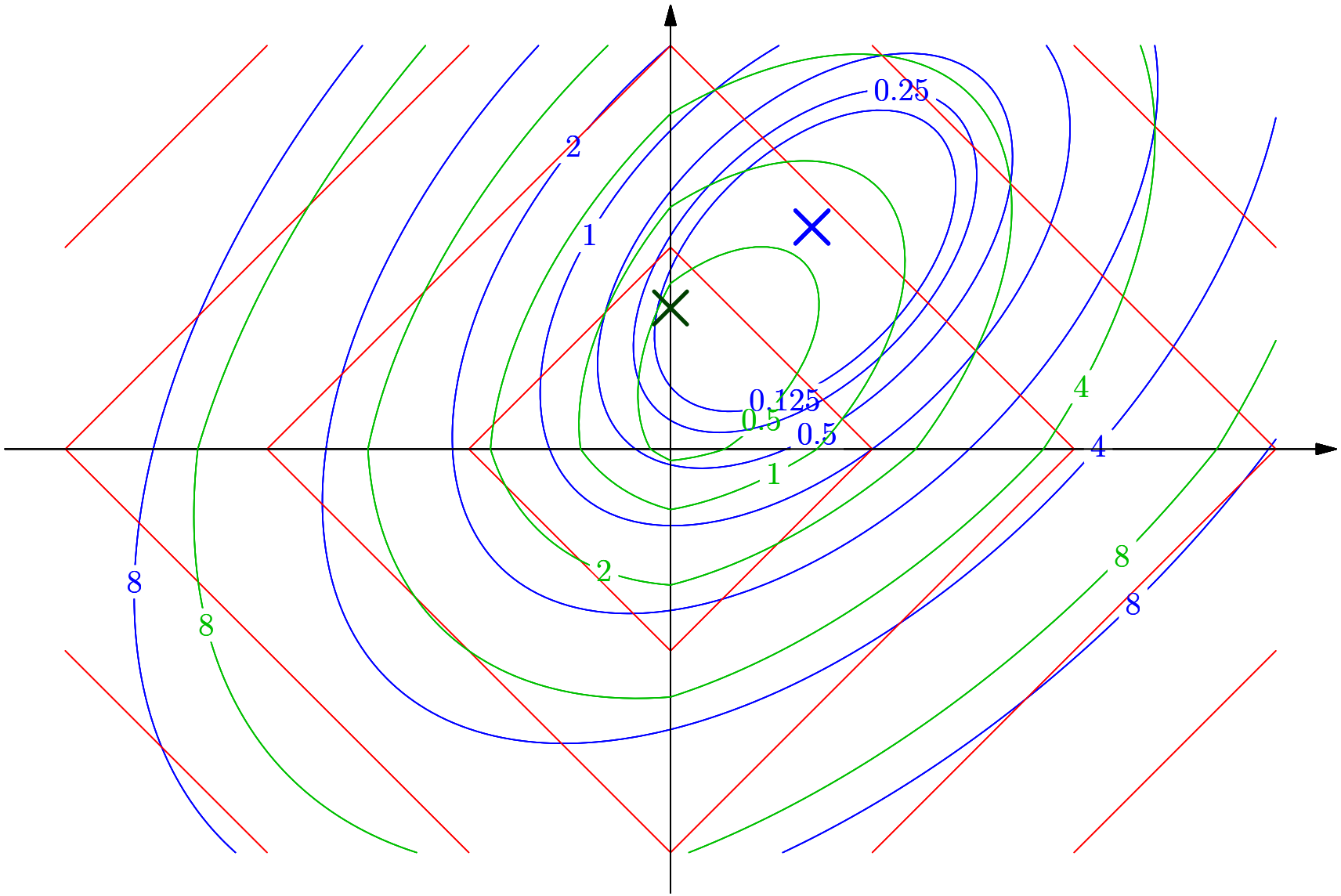
- with $\mathbf{S} = \frac{1}{T}\sum_{t=1}^{T}(\boldsymbol{X}(t) - \hat{\boldsymbol{\mu}})\,(\boldsymbol{X}(t) - \hat{\boldsymbol{\mu}})^{\mathsf{T}}$

Regularisation

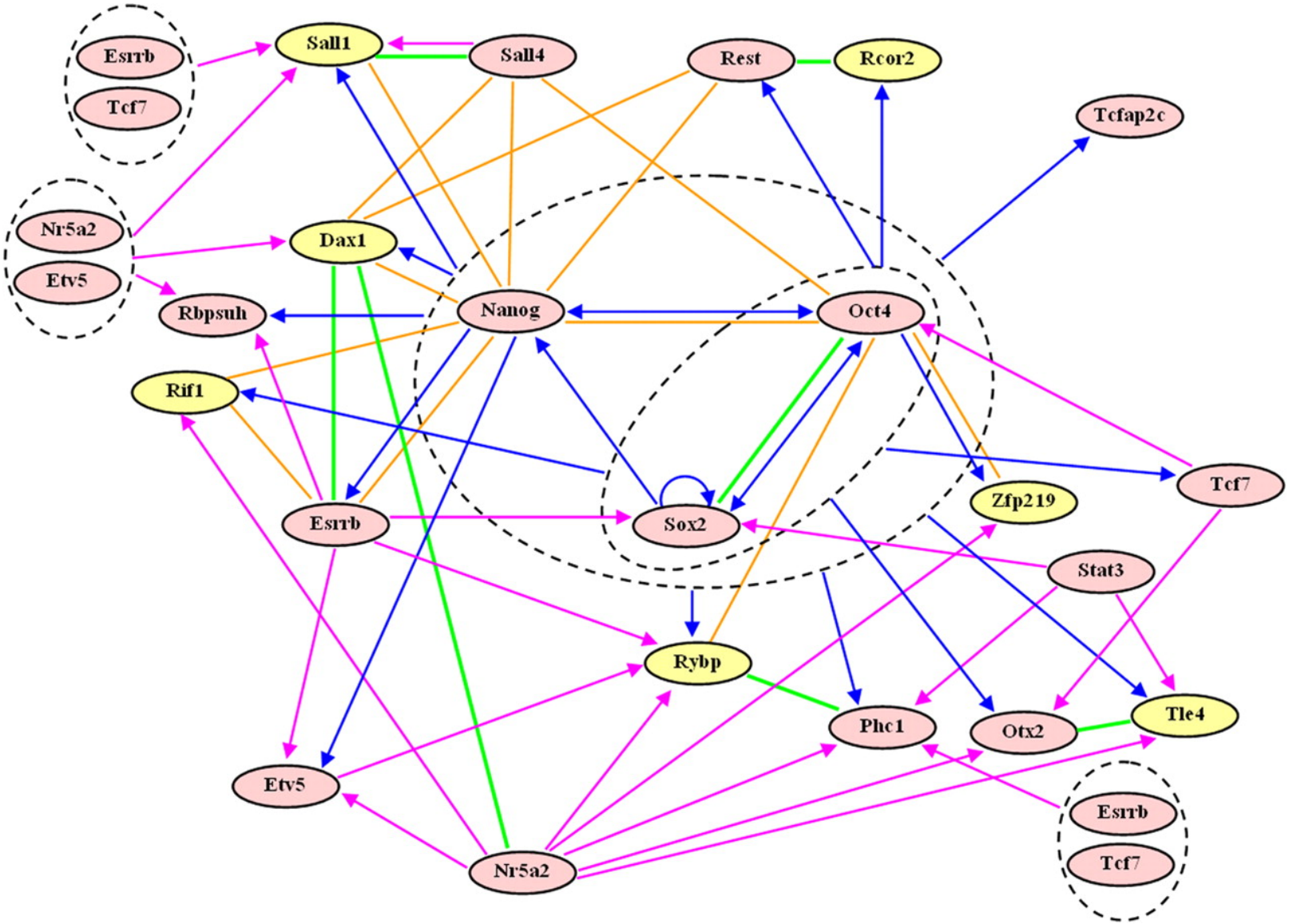
- The maximum likelihood value of the covariance is (solving for $\partial\mathcal{L}/\partial\mathbf{\Sigma} = 0$) given by $\mathbf{\Sigma} = \mathbf{S}$
- However we want to find solutions where $\mathbf{\Theta} = \mathbf{\Sigma}^{-1}$ is sparse
- Note that $|\mathbf{\Sigma}| = 1/|\mathbf{\Sigma}^{-1}| = 1/|\mathbf{\Theta}|$
- So $\mathcal{L} = \frac{T}{2}\log(|\mathbf{\Theta}|) - \frac{1}{2}\mathrm{tr}\,\mathbf{S}\,\mathbf{\Theta}$
- We can add a "regulariser" to drive small Θ_{ij} to 0

$$\mathcal{L} - \|\mathbf{\Theta}\|_1 = \frac{T}{2}\log(|\mathbf{\Theta}|) - \frac{1}{2}\mathrm{tr}\,\mathbf{S}\,\mathbf{\Theta} - \|\mathbf{\Theta}\|_1$$

Graphical Lasso



Gene Regulation Network



Summary

- We can use DNA microarrays or RNA sequencing to measure the expression of different proteins
- By measuring the expression over time we can measure correlations between proteins
- To infer a sparse graphical model that explains the observed correlations we can use the "graphical Lasso" to find a sparse inverse correlation matrix
- This is a first step to understand the control mechanisms of a cell